MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 09** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

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Q1. The wheels of a bus make 210 revolutions per minute. If the diameter of each wheel is 1 m, the speed of the bus in km/h is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 39.6 km/h (B) 19.8 km/h (C) 66 km/h (D) 33 km/h

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Distance in One Minute [0.5 Mark]

Diameter $d = 1$ m $\implies$ Circumference $= \pi d = \frac{22}{7} \times 1 = \frac{22}{7}$ m.

$$\text{Distance in 1 min} = 210 \times \frac{22}{7} = 30 \times 22 = 660 \text{ m}$$

Step 2: Speed Conversion to km/h [0.5 Mark]

$$\text{Speed} = \frac{660 \text{ m}}{1 \text{ min}} = \frac{660 \times 60}{1000} \text{ km/h} = \frac{39600}{1000} = 39.6 \text{ km/h}$$

Correct Option: (A) 39.6 km/h

$$\boxed{\text{Speed of the bus} = \mathbf{39.6} \text{ km/h}}$$

Q2. Three sectors of a circle of radius r have central angles $35°, 65°$, and $80°$. The sum of their areas equals: [1]

(A) $\frac{1}{4}\pi r^2$ (B) $\frac{1}{3}\pi r^2$ (C) $\frac{1}{2}\pi r^2$ (D) $\frac{2}{3}\pi r^2$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Combined Central Angle [0.5 Mark]

$$\theta_{\text{sum}} = 35^\circ + 65^\circ + 80^\circ = 180^\circ$$

Step 2: Sum of Areas [0.5 Mark]

$$\text{Combined Area} = \frac{180^\circ}{360^\circ} \times \pi r^2 = \frac{1}{2}\pi r^2$$

Correct Option: (C) $\frac{1}{2}\pi r^2$

$$\boxed{\text{Sum of sector areas} = \frac{\mathbf{1}}{\mathbf{2}}\pi\mathbf{r^2}}$$

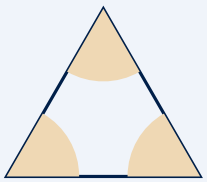
Q3. From each vertex of an equilateral triangle of side 12 cm, a sector of radius 6 cm is drawn. The sum of the areas of the three sectors is: (Use $\pi = 3.14$) [1]

(A) 56.52 cm^2 (B) 113.04 cm^2 (C) 37.68 cm^2 (D) 18.84 cm^2

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Vertex Angles of Equilateral Triangle [0.5 Mark]

Each vertex angle of the equilateral triangle is 60°.



$$\text{Total angle} = 3 \times 60^\circ = 180^\circ$$

Step 2: Calculation of Combined Area [0.5 Mark]

Radius $r = 6$ cm.

$$\text{Area} = \frac{180^\circ}{360^\circ} \times \pi r^2 = \frac{1}{2} \times 3.14 \times (6)^2 = 18 \times 3.14 = 56.52 \text{ cm}^2$$

Correct Option: (A) 56.52 cm^2

$$\boxed{\text{Sum of sector areas} = \mathbf{56.52}\ \text{cm}^\mathbf{2}}$$

Q4. A right triangle has legs 6 cm and 8 cm. The area of its incircle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 12.57 cm^2 (B) 25.14 cm^2 (C) 4π cm^2 (D) 16π cm^2

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Inradius of Right Triangle [0.5 Mark]

Hypotenuse $c = \sqrt{6^2 + 8^2} = 10$ cm.

$$r = \frac{a + b - c}{2} = \frac{6 + 8 - 10}{2} = \frac{4}{2} = 2 \text{ cm}$$

Step 2: Area of Incircle [0.5 Mark]

$$\text{Area} = \pi r^2 = \pi \times (2)^2 = 4\pi \text{ cm}^2 \approx 12.57 \text{ cm}^2$$

Correct Option: (C) 4π cm^2 [also equivalent to (A) 12.57 cm^2]

$$\boxed{\text{Area of incircle} = \mathbf{4\pi}\ \text{cm}^\mathbf{2} \quad (\approx \mathbf{12.57}\ \text{cm}^\mathbf{2})}$$

Q5. If a sector of central angle $60°$ has an area of $\frac{132}{7}$ cm^2, then the radius of the circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 6 cm (B) 7 cm (C) 5 cm (D) 8 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Formula Setup [0.5 Mark]

$$\text{Area} = \frac{60°}{360°} \times \frac{22}{7} \times r^2 = \frac{132}{7} \implies \frac{1}{6} \times \frac{22}{7} \times r^2 = \frac{132}{7}$$

Step 2: Solving for Radius r [0.5 Mark]

$$\frac{11}{21} r^2 = \frac{132}{7} \implies r^2 = \frac{132 \times 21}{7 \times 11} = 12 \times 3 = 36 \implies r = 6 \text{ cm}$$

Correct Option: **(A) 6 cm**

Radius of the circle $r = \mathbf{6}$ cm

Q6. **Assertion (A):** A car wheel covers a distance of 22 km in 5000 complete revolutions if its radius is 70 cm.
Reason (R): The distance covered by a wheel in n revolutions is given by Distance $= n \times 2\pi r$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verification of Reason (R) [0.5 Mark]

In one complete revolution, distance covered equals circumference $2\pi r$. Thus in n revolutions, distance $= n \times 2\pi r$. Reason (R) is **True**.

Step 2: Verification of Assertion (A) [0.5 Mark]

Radius $r = 70$ cm $= 0.7$ m.

$$\text{Distance} = 5000 \times 2 \times \frac{22}{7} \times 0.7 = 5000 \times 4.4 = 22000 \text{ m} = 22 \text{ km}$$

Assertion (A) is **True**, and Reason (R) correctly explains (A).

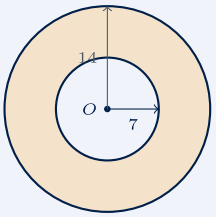
Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

Q7. Two concentric circles have radii 14 cm and 7 cm. Find the area of the circular ring bounded by them. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Outer radius $R = 14$ cm, Inner radius $r = 7$ cm.



Step 1: Formula Declaration [1 Mark]

$$\text{Area} = \pi(R^2 - r^2) = \pi(R - r)(R + r)$$

Step 2: Substitution and Calculation [1 Mark]

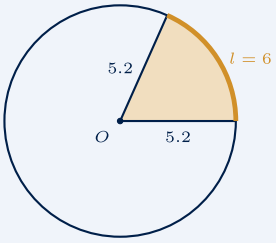
$$\text{Area} = \frac{22}{7} \times (14 - 7)(14 + 7) = \frac{22}{7} \times 7 \times 21 = 22 \times 21 = 462 \text{ cm}^2$$

Area of the circular ring = **462 cm²**

Q8. The perimeter of a sector of a circle of radius 5.2 cm is 16.4 cm. Find the area of the sector. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Radius $r = 5.2$ cm, Perimeter = 16.4 cm.



Step 1: Compute Arc Length l [1 Mark]

$$\text{Perimeter} = l + 2r = 16.4$$

$$l + 2(5.2) = 16.4 \implies l = 16.4 - 10.4 = 6 \text{ cm}$$

Step 2: Area of the Sector [1 Mark]

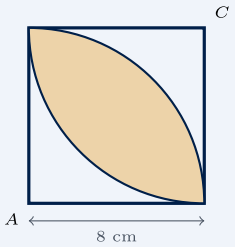
$$\text{Area} = \frac{1}{2}lr = \frac{1}{2} \times 6 \times 5.2 = 3 \times 5.2 = 15.6 \text{ cm}^2$$

Area of the sector = **15.6 cm²**

Q9. Calculate the area of the shaded leaf design between two quadrants of circles of radius 8 cm each drawn from opposite vertices of a square of side 8 cm. (Use $\pi = 3.14$) **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Side of square $a = 8$ cm. Radius of each quadrant $r = 8$ cm.



Step 1: Formula for Leaf Area [1 Mark]

$$\text{Area of leaf} = 2 \times (\text{Area of quadrant}) - \text{Area of square}$$
$$= 2 \times \left(\frac{1}{4}\pi r^2\right) - a^2 = \frac{\pi r^2}{2} - a^2$$

Step 2: Substitution with $\pi = 3.14$ [1.5 Marks]

$$\text{Area} = \frac{3.14 \times 8^2}{2} - 8^2 = \frac{3.14 \times 64}{2} - 64$$
$$= (3.14 \times 32) - 64 = 100.48 - 64 = 36.48 \text{ cm}^2$$

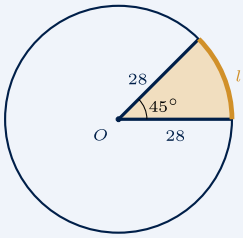
Step 3: Boxed Answer [0.5 Mark]

$$\boxed{\text{Area of shaded leaf design} = \mathbf{36.48} \text{ cm}^2}$$

Q10. In a circle of radius 28 cm, an arc subtends an angle of 45° at the centre. Find the length of the arc and the area of the sector. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Radius $r = 28$ cm, Central Angle $\theta = 45°$.



Part (i): Length of the Arc (l) [1.5 Marks]

$$l = \frac{\theta}{360°} \times 2\pi r = \frac{45°}{360°} \times 2 \times \frac{22}{7} \times 28 = \frac{1}{8} \times 176 = 22 \text{ cm}$$

Part (ii): Area of the Sector [1.5 Marks]

$$\text{Area} = \frac{\theta}{360°} \times \pi r^2 = \frac{45°}{360°} \times \frac{22}{7} \times 28 \times 28$$
$$= \frac{1}{8} \times 22 \times 4 \times 28 = \frac{1}{8} \times 2464 = 308 \text{ cm}^2$$

$$\boxed{\text{Arc Length } l = \mathbf{22} \text{ cm}, \quad \text{Sector Area} = \mathbf{308} \text{ cm}^2}$$

Q11. In a circle of radius 12 cm, a chord AB subtends an angle of 120° at the centre O.

(a) Draw altitude $OM \perp AB$ and derive the exact formula for the area of $\triangle OAB$ in terms of radius r and angle 120°. **[2]**

(b) Calculate the area of the minor segment bounded by chord AB. **[2]**

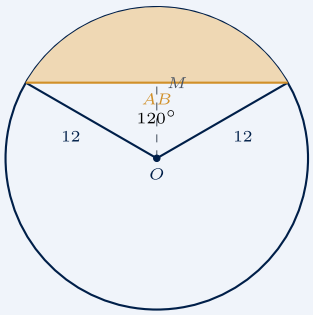
(c) Calculate the area of the corresponding major segment. (Use $\pi = 3.14, \sqrt{3} = 1.73$) **[1]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Radius $r = 12$ cm, Central Angle $\theta = 120^\circ$.

Part (a): Derivation of Area of $\triangle OAB$ [2 Marks]

Draw $OM \perp AB$. In isosceles $\triangle OAB$, OM bisects $\angle AOB \implies \angle AOM = 60^\circ$ and $AM = MB$.



$$OM = r\cos 60^\circ = \frac{r}{2}, \quad AM = r\sin 60^\circ = \frac{r\sqrt{3}}{2} \implies AB = r\sqrt{3}$$

$$\text{Area}(\triangle OAB) = \frac{1}{2} \times AB \times OM = \frac{1}{2} \times r\sqrt{3} \times \frac{r}{2} = \frac{\sqrt{3}}{4}r^2$$

$$\text{Area} = \frac{\sqrt{3}}{4} \times (12)^2 = 36\sqrt{3} = 36 \times 1.73 = 62.28 \text{ cm}^2$$

Part (b): Area of Minor Segment [2 Marks]

$$\text{Area of minor sector} = \frac{120^\circ}{360^\circ} \times 3.14 \times (12)^2 = \frac{1}{3} \times 452.16 = 150.72 \text{ cm}^2$$

$$\text{Area of minor segment} = 150.72 - 62.28 = 88.44 \text{ cm}^2$$

Part (c): Area of Major Segment [1 Mark]

$$\text{Total Circle Area} = 3.14 \times (12)^2 = 452.16 \text{ cm}^2$$

$$\text{Area of major segment} = 452.16 - 88.44 = 363.72 \text{ cm}^2$$

(a) Area($\triangle OAB$) = **62.28 cm²**, (b) Minor = **88.44 cm²**, (c) Major = **363.72 cm²**

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Q12. A designer creates nine identical circular floral motifs of radius 7 cm on a square silk handkerchief of side 42 cm.

(a) Find the total circumference of the nine circular motifs. **[1]**

(b) Find the total area occupied by the nine circular motifs. **[1.5]**

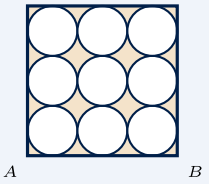
(c) Find the area of the remaining silk handkerchief. **[1.5]**

(d) If the remaining portion is dyed at the rate of Rs. 0.15 per cm^2, find the total dyeing cost. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[1]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Given: Side of square $a = 42$ cm, Radius of each motif $r = 7$ cm.

Part (a): Total Circumference of 9 Motifs [1 Mark]



$$\text{Circumference of 1 motif} = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

$$\text{Total Circumference} = 9 \times 44 = 396 \text{ cm}$$

Part (b): Total Area of 9 Motifs [1.5 Marks]

$$\text{Area of 1 motif} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

$$\text{Total Area} = 9 \times 154 = 1386 \text{ cm}^2$$

Part (c): Remaining Area of Handkerchief [1.5 Marks]

$$\text{Total Area of square} = 42 \times 42 = 1764 \text{ cm}^2$$

$$\text{Remaining Area} = 1764 - 1386 = 378 \text{ cm}^2$$

Part (d): Total Dyeing Cost [1 Mark]

$$\text{Cost} = 378 \times \text{Rs. } 0.15 = \text{Rs. } 56.70$$

(a) **396** cm, (b) **1386** cm^2, (c) Remaining = **378** cm^2, (d) Cost = Rs. **56.70**

SECTION E: Case-Based / Applied Situation

MATHS MANTHAN | $(1 \times 4 = 4 \text{ Marks})$

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Q13. Case Study: Maritime Lighthouse Harbor Warning Sector

A coastal port authority maintains a powerful lighthouse situated on a rocky promontory. The lighthouse beacon projects an intense red warning beam through a sector angle of 80° across the sea surface to a maximum distance of 16.5 km to steer vessels away from shallow reefs.

(i) State the geometric relationship between the sector radius and the light range. [1]

(ii) Calculate the length of the sea boundary arc illuminated by the warning beam. [1]

(iii) Calculate the total area of the sea over which vessels receive navigation warning. (Use $\pi = 3.14$) [2]

— OR —

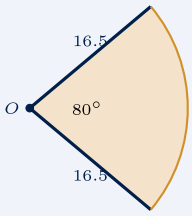
If the light range is increased to 20 km, find the new warning coverage area. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Initial range $r = 16.5$ km, Sector angle $\theta = 80^\circ$.

Part (i): Geometric Relationship [1 Mark]

The maximum range of the lighthouse beam defines the **radius** (r) of the illuminated circular sector: $r = 16.5$ km.



Part (ii): Length of the Sea Boundary Arc [1 Mark]

$$l = \frac{80^\circ}{360^\circ} \times 2\pi r = \frac{2}{9} \times 2 \times 3.14 \times 16.5 = \frac{207.24}{9} \approx 23.03 \text{ km}$$

Part (iii): Warning Coverage Area ($r = 16.5$ km) [2 Marks]

$$\text{Area} = \frac{80^\circ}{360^\circ} \times 3.14 \times (16.5)^2 = \frac{2}{9} \times 3.14 \times 272.25 = \frac{1709.73}{9} = 189.97 \text{ km}^2$$

— OR (Alternative Choice) —

New Coverage Area with $r' = 20$ km: [2 Marks]

$$\text{New Area} = \frac{80^\circ}{360^\circ} \times 3.14 \times (20)^2 = \frac{2}{9} \times 3.14 \times 400 = \frac{2512}{9} \approx 279.11 \text{ km}^2$$

(i) Radius r = Range, (ii) Arc = **23.03** km, (iii) Area = **189.97 km²** (OR New Area = **279.11 km²**)

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